SlideChat: A Large Vision-Language Assistant for Whole-Slide Pathology Image Understanding

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Abstract

Despite the progress made by multimodal large language models (MLLMs) in computational pathology, they remain limited by a predominant focus on patch-level analysis, missing essential contextual information at the whole-slide level. The lack of large-scale instruction datasets and the gigapixel scale of whole slide images (WSIs) pose significant developmental challenges. In this paper, we present SlideChat, the first vision-language assistant capable of understanding gigapixel whole-slide images, exhibiting excellent multimodal conversational capability and response complex instruction across diverse pathology scenarios. To support its development, we created Slide-Instruction, the largest instruction-following dataset for WSIs consisting of 4.2K WSI captions and 176K VQA pairs with multiple categories. Furthermore, we propose SlideBench, a multimodal benchmark that incorporates captioning and VQA tasks to assess SlideChat's capabilities in various settings such as microscopy, diagnosis and clinical. Compared to both general and specialized MLLMs, SlideChat exhibits exceptional capabilities, achieving state-of-the-art performance on 18 of 22 tasks. For example, it achieved an overall accuracy of 81.17% on SlideBench-VQA (TCGA), and 54.15% on SlideBench-VQA (BCNB). Our code, data, and model is publicly accessible at https://uni-medical.github.io/SlideChat.github.io.

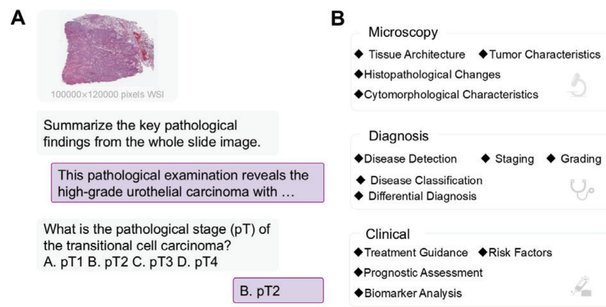

Figure 1. SlideChat is the first large vision-language assistant specifically designed for whole-slide pathology analysis. SlideChat can generate comprehensive descriptions of whole-slide images and provide contextually relevant responses across various applications.

1. Introduction

Computational pathology aims to improve the analysis of digitized tissue samples [14, 24], such as whole slide images (WSIs), by applying artificial intelligence to aid in the diagnosis, identification, and understanding of disease. Recently, the development of this field has gained rapid momentum, mainly driven by breakthroughs in the visual foundation model [9, 27, 30]. These models learn generalized representations by pre-training on large-scale data and perform well in various downstream tasks, such as cancer detection and biomarker prediction. Building on this base, integration with the powerful Large Language Models (LLMs) further advances the development of the Multimodal Large Language Model (MLLMs) [20], which has made great strides in responding to more complex, open-ended visual queries, enabling it to serve as a versatile assistant at various stages of medical care, including clinical decision-making, education, and research.

Nevertheless, there are some major challenges that hinder the development and use of pathology MLLMs for real-world applications. First, it is challenging to develop a MLLM architecture that can effectively be capable of gigapixel whole slides (*e.g.*, 100,000 × 100,000 pixels). Existing models [20, 22, 26] often process whole slides by extracting small patch/ROI-level data for subsequent analysis, resulting in a limited understanding of global slide con-

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text and suboptimal performance in some complex pathological analysis. Second, publicly available multi-modal pathology slide datasets are relatively scarce and of varying quality [7, 8, 12], which limits the development of MLLMs trained on such data.

In this paper, we present SlideChat (see Fig. 1), the first open-source vision-language assistant capable of understanding gigapixel whole-slide images. First, SlideChat is trained on SlideInstruction, a large-scale multi-modal instruction dataset encompassing data from The Cancer Genome Atlas (TCGA) [15] via our specifically designed data processing pipeline (see Fig. 3 (A)). SlideInstruction contains 4,181 WSI-caption pairs and 175,753 visual question-answer pairs from 3,294 patients, covering 10 cancer types. The question-answer pairs include both open-ended and closed-ended questions, further divided into 13 subcategories, covering a diverse range of clinical tasks. SlideInstruction is more than 20 times larger than previous public instruction datasets in the number of instructions. Second, we propose SlideChat, a novel architecture in LLaVA style for capable multi-modal analysis of gigapixel whole slides. A gigapixel whole slide is first divided into a series of patches, each of which is individually processed by a patch-level encoder to extract local features. The resulting long sequence of patch tokens is then processed by a slide-level encoder employing sparse attention to enhance the slide-level features. Finally, enhanced patch tokens are fed into the LLM via a projector, which processes user queries and generates responses.

To systematically evaluate the performance of SlideChat in real-world scenarios, we establish a comprehensive digital pathology benchmark named SlideBench, encompassing more than 20 clinical tasks, using data from both TCGA and the in-the-wild Early Breast Cancer Core-Needle Biopsy (BCNB) dataset. This resulted in three test sets: SlideBench-Caption, comprising 734 WSI-caption pairs; SlideBench-VQA (TCGA), comprising 7,827 VQA pairs covering 13 different tasks; and SlideBench-VQA (BCNB), including a total of 7,274 VQA entries from 1,058 patients, covering seven different tasks. Additionally, we compare the performance of SlideChat on another externally proposed dataset, WSI-VQA [8], to further validate its effectiveness. We compare our model with the currently available state-of-the-art general and medical-specialized MLLMs including GPT-4o, LLava-Med [16], MedDr [13] and Quilt-LLaVA [22]. Benefiting from large-scale, high-quality training and effective local-global context modeling, SlideChat achieves state-of-the-art performance on 18 out of 22 tasks, with significant improvements over the second-best method 10% on 9 tasks on four benchmarks. Specifically, SlideChat achieves an average accuracy improvement of 13.47% over the second-best model on SlideBench-VQA (TCGA), an average improvement of 12.59% on SlideBench-VQA (BCNB), and an improvement of 5.82% on WSI-VQA. Finally, to accelerate research progress in digital pathology, we make SlideChat fully open-weight, including source code and model weights as well as instruction and benchmark datasets. The key contributions are summarized four-fold in the following:

- We develop SlideChat, the first vision-language assistant capable of understanding gigapixel whole-slide images, achieving state-of-the-art performance on multiple benchmarks.
- We create SlideInstruction, a largest comprehensive WSI instruction-following dataset containing 4.2K WSI-caption pairs and 176K VQA pairs.
- We establish SlideBench, a WSI multi-modal benchmark comprising SlideBench-Caption, SlideBench-VQA (TCGA), and SlideBench-VQA (BCNB), covering 21 different clinical tasks.
- We release SlideChat, SlideInstruction and SlideBench as open-source resources to facilitate research and development in computational pathology.

2. Related Works

Whole Slide Image Analysis Whole slide images are pivotal in modern pathology, enabling comprehensive analysis of tissue samples for tasks such as predicting patient prognosis, classifying cancer subtypes, and identifying biomarkers [17, 18, 23–25, 33]. Recent studies have leveraged pathology foundational models [3, 28, 31] to enhance WSIs analysis, either through fine-tuning for specific downstream tasks or by employing zero-shot prediction approaches in CLIP [21] style. Although these models are effective in task-specific applications, their reliance on fine-tuning or limited zero-shot capabilities restricts their generalizability across diverse and complex user instructions.

MLLMs in Computational Pathology The paradigm of MLLMs enables them to effectively respond to more complex, open-ended visual queries while processing pathology images, thus providing significant value across various medical stages. PathChat [20] is a vision-language assistant designed for pathology, developed with 450K private instruction pairs to handle both visual and natural language queries. Quilt-Instruct [22] is a large-scale dataset comprising 107K question-answer pairs. Building on Quilt-Instruct, Quilt-LLAVA [22] is a model designed for diagnostic reasoning across multiple image patches, leveraging its extensive question-answer pairs to accurately interpret complex H&E images. PathAsst [26] combines a pathology-specific CLIP model with Vicuna-13b [10] to create a multimodal generative foundational model tailored for pathology. However, current MLLMs primarily only focus on patch or ROI data, limiting their utility for slide-level clinical applications where broader contextual understanding is crucial.

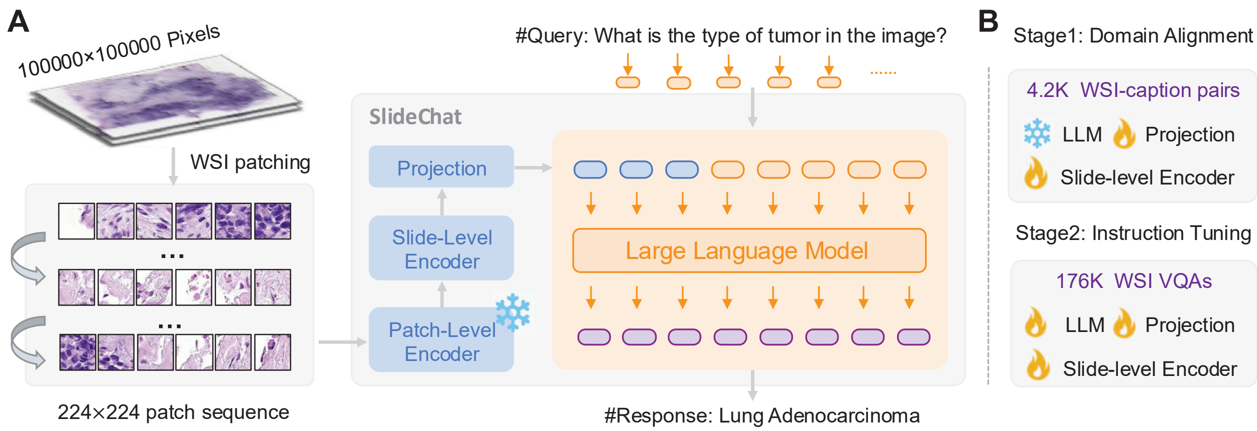

Figure 2. Overview of our SlideChat. (A) SlideChat serializes each input WSI into a sequence of 224×224 patches, converting each into visual embeddings with a patch-level encoder. A slide-level encoder then interacts with these features to generate contextual embeddings. Then, a multimodal projector maps the visual features from the slide-level encoder into a unified space, aligned seamlessly with the LLM. (B) SlideChat was trained for two stages: Cross-Domain Alignment and Visual Instruction Learning.

3. SlideChat

3.1. Architecture

To achieve the goal of analyzing gigapixel whole-slide images in a multimodal setting, as shown in Fig. 2, SlideChat consists of four key designs: the patch-level encoder, the slide-level encoder, the multimodal projector module, and the large language model. Our method starts by partitioning the WSI into smaller 224 × 224 pixel patches, making it computationally feasible to process such large images. These patches are then passed through a well-trained, frozen patch-level encoder [19], which extracts localized features from each individual patch, capturing fine-grained details such as cellular structures. Building on this, we employ LongNet [11, 30] as slide-level encoder to enhance the patch-level embeddings and capture global patterns across the entire slide. This encoder uses sparse attention mechanisms to learn both local and global contextual information, enabling the model to perceive intricate local features while capturing the broader context, which is critical for comprehensive pathological assessments. Following the slide-level encoding, SlideChat incorporates a multimodal projector that maps these patch tokens into a unified space aligned with the LLM. This ensures that the visual features extracted from the WSIs are effectively transformed into representations compatible with the language model, facilitating seamless integration and interaction between visual and textual data. Concurrently, the model accepts natural language instructions from users, such as "What is the type of tumor in the image?". These textual queries are processed by the LLM, which comprehends the textual input and integrates it with the visual features extracted from the WSIs, enabling accurate and contextually relevant diagnostic responses. This multimodal reasoning capability allows SlideChat to provide accurate and contextually relevant answers to complex pathology-related questions, thereby supporting clinical decision-making, education, and research across various medical stages.

3.2. Data

SlideInstruction There is a notable lack of large-scale multimodal pathology datasets supporting the training of vision-language assistants for whole-slide image understanding. To support the training of SlideChat, we develop SlideInstruction, a comprehensive instruction dataset, sourced from the TCGA database, comprising 4,915 WSI-report pairs from 4,028 patients. Fig. 3 illustrates our entire data curation pipeine. We initially prompt GPT-4 to refine the pathology reports, and clean up the noise in the report including *unrelated symbols, technical details of pathology department procedures, specimen handling and processing information, redundant administrative or legal statements, and some repeated information*. For the refined pathology reports, we further employ GPT-4 to generate high-quality multimodal data, comprising two main components: (1) *WSI-Caption Data*: We craft concise, clinically relevant summaries for each whole slide image by prompting the language model to extract key pathological findings. These summaries were structured into coherent paragraphs that highlighted crucial clinical details such as diagnostic results, tumor characteristics, margin status, and lymph node involvement, ensuring the caption dataset is both focused and informative. (2) *WSI Instruction-Following Data*: To enhance the model's ability to follow instructions and improve its comprehension of pathology images, we leveraged GPT-4 to generate tailored question-

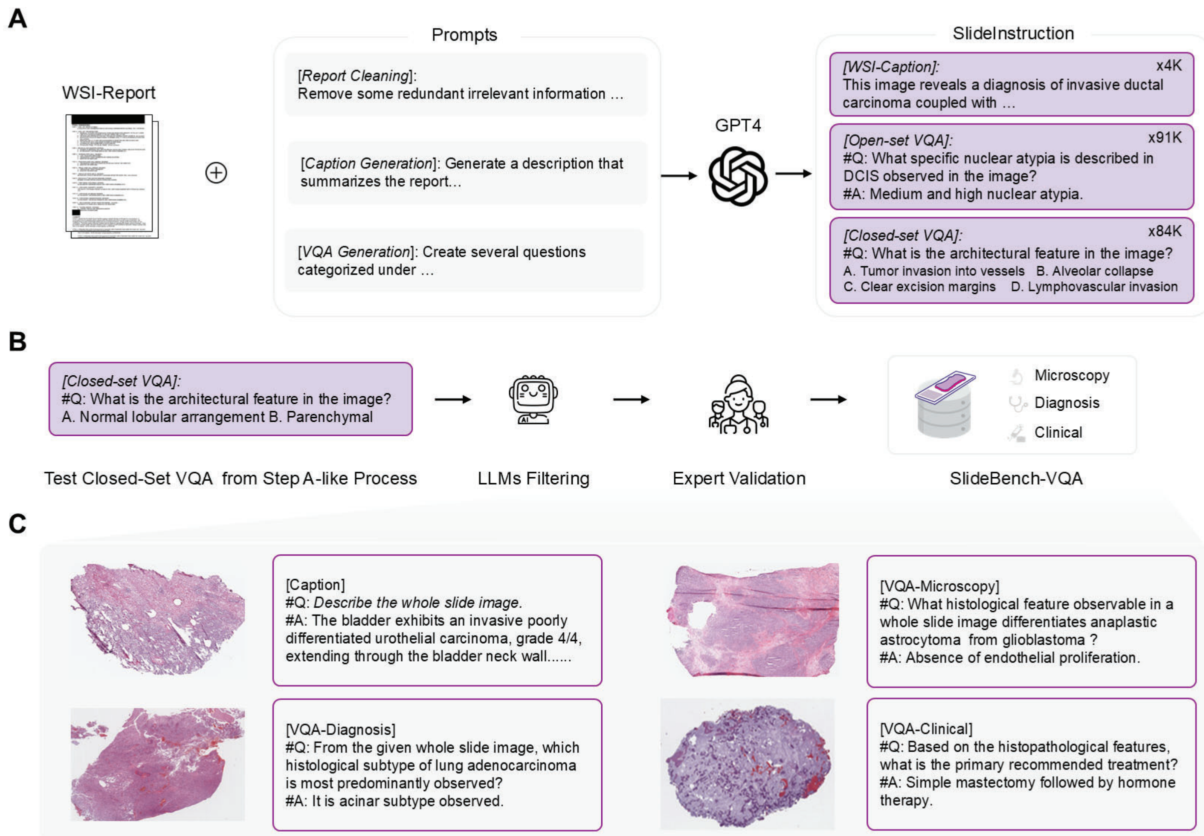

Figure 3. (A) Overview of the SlideInstruction generation pipeline. We prompt GPT-4 to extract the WSI-Caption, Open-set VQA and Closed-set VQA from reports. (B) For the generated Closed-set VQA, we employ LLMs to filter low-quality QA pairs and involve pathologists for validation, resulting in the creation of SlideBench-VQA. (C) Examples of WSI caption and instruction-following scenarios in microscopy, diagnostics, and clinical applications. For additional examples, please refer to Supplementary Material.

and-answer pairs for each WSI report. Drawing inspiration from PathChat [20], we structure these questions into 3 broad categories—microscopy, diagnosis, and clinical considerations—which represent key stages in the pathology workflow, and 13 narrow categories focusing on specific aspects within each stage (Fig. 1 B). To construct a comprehensive instructional dataset, we began by generating two open-ended and two closed-ended QA pairs for each report within each specific category, taking advantage of the rich content found in the majority of TCGA reports. Then, some QA pairs were excluded, such as those lacking corresponding category information. Regarding the train/test split, it is worth noting that the WSI-report datasets from TCGA include 2 types: (a) one report linked to multiple WSIs, and (b) one report linked to a single WSI. For type (a), where specific diagnostic details may not align perfectly with each WSI, we include all WSIs in the training set to introduce some "noisy data", which can enhance model robustness. For type (b), 80% of WSIs are allocated to the training set and 20% to the test set. Finally, there are 4,181 WSIs for training and 734 WSIs for testing. Consequently, we construct a large-scale training set named SlideInstruction, comprising 4,181 WSI captions and 175,753 instruction-following VQA pairs across 13 narrow categories.

SlideBench To systematically evaluate the performance of SlideChat, we include the remaining 734 WSI captions from the test set, along with a substantial number of closed-set VQA pairs, to establish a comprehensive evaluation benchmark. First, we construct a set named SlideBench-Caption based on the WSI-Caption data to evaluate the model's ability to generate accurate and coherent descriptions of whole slide images. Secondly, we construct SlideBench-VQA (TCGA) based on closed-set VQA pairs along with test WSIs, aiming to evaluate various aspects of model performance. As shown in Fig. 3 (B), to improve the quality of the testing benchmarks, we employ four advanced large language models, including GPT-4 [2], InternLM2-

Chat-7B [6], Qwen-7B-Chat [5], and DeepSeek-7B-Chat, to filter closed-set VQAs by predicting answers based solely on the question text. Any questions for which at least three of these models provided correct answers are subsequently excluded. Following this automated filtering, five expert pathologists are invited to review and amend the remaining questions. The review process is guided by the following criteria: (1) Whether the correct answer necessitates image interpretation; (2) Whether the question and its corresponding answer are logically and coherently structured; and (3) Whether the question aligns appropriately with the designated broad and narrow categories. VQA pairs failing to meet these criteria are excluded by the pathologists. Consequently, the SlideBench-VQA (TCGA) comprises 7,827 VQAs across 13 categories, with some examples illustrated in Fig. 3 C. Additionally, we incorporate the in-the-wild Early Breast Cancer Core-Needle Biopsy (BCNB) WSI dataset [29], which encompasses a diverse patient population and a variety of clinical task labels, to enhance the test set benchmark and more comprehensively assess the model's generalization capabilities. In detail, we convert the BCNB dataset into a VQA format by rephrasing the classification objectives into a specific template as questions, while transforming the original multi-class labels into selectable options, and integrating it into SlideBench as an external subset, named SlideBench-VQA (BCNB). This dataset comprises 7,247 VQA pairs from 1,058 patients, specifically designed to evaluate SlideChat's zero-shot generalization capability across 7 distinct classification tasks.

3.3. Two-stage Training

Stage 1: Cross-Domain Alignment. SlideChat adopts a two-stage training approach (see Fig. 2 B). In the first stage, the primary objective is to align the LLM's word embeddings with the visual features extracted from whole slide images. This alignment enables the LLM to interpret visual representations from the slide-level encoder, facilitating the effective utilization of the intricate features within the slides. During this stage, SlideChat is trained to generate descriptive captions using 4.2K WSI-caption pairs from SlideInstruction. Specifically, only the slide-level encoder and projection matrix are updated, while the patch-level encoder and LLM weights remain fixed.

Stage 2: Visual Instruction Learning. In the second stage, we focus on visual question-answering tasks to train the model to accurately respond to domain-specific questions concerning whole slide images. During this phase, the model develops the ability to handle a broad range of multimodal instructions, enabling it to generate answers by effectively integrating both visual and textual information. For example, the model must perform various pathology tasks, such as describing the extent of tumor invasion or assessing the degree of cellular differentiation. To accomplish this, we utilize 176K WSI VQAs from SlideInstruction in the second training stage, allowing the slide encoder, projection layer, and large language model components to be fully trainable to ensure comprehensive adaptability. This approach significantly enhances the model's capability to handle diverse pathology-related tasks, thereby increasing its effectiveness in real-world clinical and research settings.

4. Experiment

We conducted the following experiments to evaluate three key aspects of SlideChat: (1) its whole slide image captioning capability, which assesses proficiency in generating descriptive captions that accurately summarize the critical pathological features of a WSI; (2) its visual question-answering (VQA) ability across various complex pathological scenarios and its generalizability in zero-shot settings; and (3) SlideChat's ability to process gigapixel WSIs, capturing both essential global context and intricate details, thereby enhancing its performance compared to patch-level MLLMs. For WSI captioning baselines, we benchmark against MI-Gen [7], a state-of-the-art method specifically designed for this task. Given that existing MLLMs cannot handle the gigapixel scale of whole slide images, we establish baseline comparisons using two approaches: (1) randomly selecting 30 patches from each WSI and inputting them into MLLMs (e.g., GPT-4o [2], LLaVA-Med [16], MedDr [16], Quilt-LLaVA [22]), followed by a majority voting scheme to generate slide-level predictions; and (2) directly inputting a WSI thumbnail, resized to 1024×1024 pixels, referred to as Slide (T), into the MLLMs. For VQA tasks, we further evaluate performance by comparing against random prediction baselines and text-only models, thereby assessing the incremental contribution of visual information. Unless otherwise specified, SlideChat is configured with the patch-level encoder CONCH [19], the slide-level encoder LongNet [11], and utilizes the Qwen2.5-7B-Instruct [32] as LLM.

Table 1. Captioning performance across different methods on SlideBench-Caption.

Methods	Input	BLEU-1	BLEU-2	BLEU-3	BLEU-4	Rouge-L	GPT score
GPT-4o	Patch	0.16	0.03	0.01	0.01	0.13	1.54
GPT-4o	Slide (T)	0.10	0.03	0.01	0.01	0.11	1.03
Quilt-LLaVA	Patch	0.30	0.19	0.11	0.05	0.18	3.87
Quilt-LLaVA	Slide (T)	0.23	0.09	0.04	0.01	0.16	1.89
MI-Gen	Slide	0.37	0.24	0.15	0.10	0.25	4.14
SlideChat	Slide	0.37	0.21	0.12	0.08	0.24	4.11

SlideBench-Caption We report BLEU, ROUGE, and GPT scores to evaluate caption generation performance in Tab. 1. For the GPT score, we use GPT-4 to assess the

Table 2. VQA performance with different methods.

Methods	Input	SlideBench-VQA (TCGA)				SlideBench-VQA (BCNB)	WSI-VQA*
		Microscopy	Diagnosis	Clinical	Overall		
Random	Text	24.44	24.91	26.44	25.02	24.40	24.14
GPT-4		38.28	29.09	45.00	37.25	0	18.60
GPT-4o	Patch	62.89	46.69	66.77	57.91	41.43	30.41
MedDr		73.30	57.78	74.25	67.70	33.67	54.36
LLaVA-Med		47.34	32.78	47.96	42.00	30.1	26.31
Quilt-LLaVA		57.76	35.96	53.07	48.07	32.19	44.43
GPT-4o	Slide (T)	38.28	23.10	43.42	34.07	0	14.03
MedDr		70.48	52.47	72.80	64.25	35.48	50.95
LLaVA-Med		45.82	27.58	40.84	37.39	0	18.79
Quilt-LLaVA		49.12	26.97	44.75	39.39	41.55	35.40
SlideChat	Slide	87.64 (+14.34)	73.27 (+15.49)	84.26 (+10.01)	81.17 (+13.47)	54.14 (+12.59)	60.18 (+5.82)

similarity between the generated captions and the ground truth, providing an overall score on a scale of 1 to 10, with higher scores indicating better performance. When utilizing patch-level inputs, GPT-4o generates individual descriptions for each patch, which are subsequently integrated to create the final slide-level caption. However, this approach yields poor performance, as evidenced by a BLEU-1 score of 0.16 and a GPT-score of 1.54. These results suggest that the patch-based method fails to adequately capture the broader context necessary for accurate WSI captioning. When the WSI thumbnail of size 1024×1024 pixels is used as input to GPT-4o, performance decreases further, with a BLEU-1 score of 0.10 and a GPT-score of 1.03. This suggests that while the thumbnail offers a global view of the slide, it may lack the resolution and detail necessary for generating precise and informative captions. In contrast, MI-Gen, a model specifically designed for WSI captioning, demonstrates significantly superior performance across all metrics, achieving a BLEU-1 score of 0.37 and a GPT score of 4.14. Similarly, SlideChat, shows competitive results with a BLEU-1 score of 0.37 and a GPT score of 4.11. These outcomes highlight SlideChat's ability to effectively integrate both local and global information from the slides and confirm its efficiency in describing whole-slide images.

SlideBench-VQA (TCGA) We further evaluate SlideChat's overall performance on the multiple-choice VQA benchmark. The results on SlideBench-VQA (TCGA) in Tab. 2, compare different methods across three categories: microscopy, diagnosis, and clinical, along with an overall performance score. Random selection achieves an overall score of 25.02% accuracy, serving as a baseline for answer distribution but demonstrating poor performance. While GPT-4, relying solely on text input, outperforms random predictions, it continues to struggle with accurately answering questions. When GPT-4o incorporates patch-level inputs, its performance improves markedly, reaching a score of 57.91% and underscoring the crucial role of detailed visual data. However, using a WSI thumbnail results in a lower score of 34.07%, as the reduced detail restricts its ability to deliver precise answers. MedDr performs well, achieving an overall score of 67.70% with patch-level inputs, though this drops slightly to 64.25% when using the slide thumbnail due to the loss of fine visual details. SlideChat outperforms all other methods, attaining a leading overall accuracy of 81.17%, excelling across all categories and significantly surpassing the competition. Even in more fine-grained pathological scenarios, as depicted in the left portion of Fig. 4, SlideChat remains the top-performing model across 13 tasks, particularly in areas such as cytomorphological characteristics, histopathological changes, disease detection, disease classification, and staging and grading, which require the identification of complex pathological visual features. Compared with baselines taking some patches or slide thumbnail as inputs, SlideChat has the capability to analyze a significantly greater number of pathological features with enhanced detail, effectively capturing both localized features and overarching global patterns, allowing SlideChat to provide more accurate and nuanced insights into pathological variations.

SlideBench-VQA (BCNB) In the zero-shot VQA setting, SlideChat significantly outperforms all other models, achieving the highest overall score of 54.14% on SlideBench-VQA (BCNB). It particularly excels in identifying tumor types, far surpassing other baselines in this task. This performance highlights SlideChat's generalization capability across a wide range of tasks. When taking patches as inputs, GPT-4o outperforms both MedDr and LLaVA-

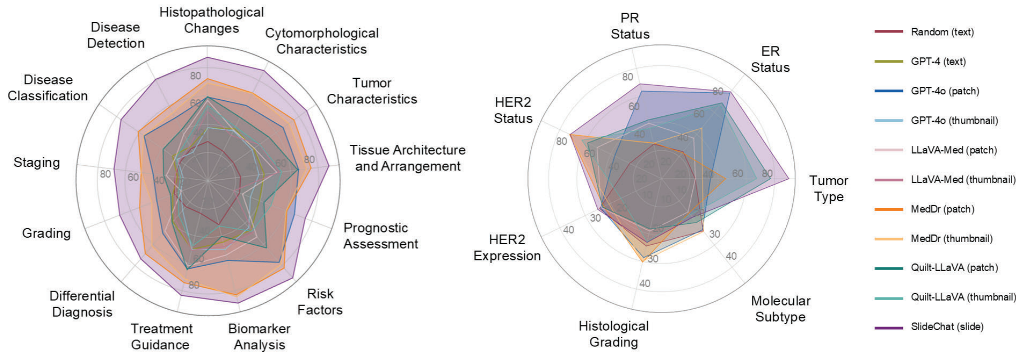

Figure 4. Accuracy on different tasks in SlideBench-VQA (TCGA) (left) and SlideBench-VQA (BCNB) (right).

Med, achieving a score of 41.43%, though it still falls short of SlideChat by 12.59%. Notably, GPT-4o and LLaVA-Med performed very poorly, achieving a score of zero across all tasks when evaluated using slide thumbnails from this testing set. MedDr and Quilt-LLaVA both achieve moderate results, with MedDr scoring 33.67% on patches and 35.48% on thumbnails, while Quilt-LLaVA attains 32.19% on patches and 41.55% on thumbnails. This outcome highlights that, for complex WSIs, relying solely on relatively sufficient visual features is inadequate for effectively supporting a range of tasks. In the more fine-grained pathological tasks of the BCNB benchmark, SlideChat attains state-of-the-art performance on 5 out of 7 tasks (Fig. 4), further demonstrating the effectiveness of SlideChat.

WSI-VQA* We also curated a subset of closed-set VQA pairs from the public WSI-VQA [8] dataset, referred to as WSI-VQA*, which exclusively consists of samples that were not in the training set, to evaluate the model's performance. SlideChat demonstrates the highest performance with a score of 60.18. Although MedDr performs well with both patch inputs (54.36%) and slide thumbnail inputs (50.95%), it still falls short compared to SlideChat. GPT-4o struggles significantly, especially with slide thumbnails, scoring only 14.03%, which highlights the limitations of using lower-resolution inputs. SlideChat leverages both detailed local information and broader context to process WSIs, making it the most effective model for this benchmark. This further emphasizes its superior capability in handling whole-slide data for VQA tasks.

Ablation We performed ablation experiments from different perspectives as follows: a) *Large Language Model Comparison*: Firstly, we compare the performance of several large language models, each with a parameter scale of approximately 7 billion, including Vicuna-7B-v1.5 [10], Phi-3-Mini-4k-Instruct [1], Qwen1.5-7B-Chat [5], Llama3-8B-Instruct [4], InternLM2-Chat-7B [6], and Qwen2.5-7B-Instruct [32]. Specifically, we measured their performance on the SlideBench-Caption using the GPT-score and their accuracy on three VQA datasets. As shown in Tab. 3, the results demonstrate that SlideChat, powered by the Qwen2.5-7B-Instruct model, achieved the highest performance across all tasks, particularly excelling in the captioning task. These findings underscore the significant potential of developing SlideChat with the Qwen2.5-7B-Instruct model. Utilizing the Qwen2.5 model, we further evaluated models of varying scales and discovered that larger models generally exhibited superior performance, particularly the 7B and 14B parameter models. While these two models showed comparable performance across the three benchmarks, the 14B model surpassed the 7B model by 2.57% on the SlideBench-VQA (TCGA). Given computational resource constraints, SlideChat uses the 7B model by default to achieve the best balance between performance and resource efficiency. b) *Slide-level encoder effectiveness*: We investigate the effectiveness of the slide-level encoder by initially removing it from SlideChat and employing a two-stage training approach. In the first stage, we exclusively trained the projection layers. However, this approach failed to reduce training loss or generate coherent text effectively,

Table 3. Performance Comparison of LLMs and Slide Encoder on WSI Captioning and VQA Tasks.

LLMs	Slide Encoder	Caption	VQA (TCGA)	VQA (BCNB)	WSI-VQA*
Vicuna-7B-v1.5	✓	3.28	41.43	41.43	31.98
Phi-3-Mini-4k-Instruct	✓	2.66	79.93	43.92	60.18
Qwen1.5-7B-Chat	✓	2.92	77.63	44.07	56.89
Llama3-8B-Instruct	✓	3.30	78.78	42.82	55.25
Internlm2-Chat-7B	✓	3.30	79.10	52.13	56.76
Qwen2.5-3B-Instruct	✓	3.40	80.32	45.79	56.38
Qwen2.5-14B-Instruct	✓	3.39	82.14	51.57	60.94
Qwen2.5-7B-Instruct	✓	4.11	81.17	54.14	60.18
Qwen2.5-7B-Instruct	×	3.95	81.21	45.49	59.67

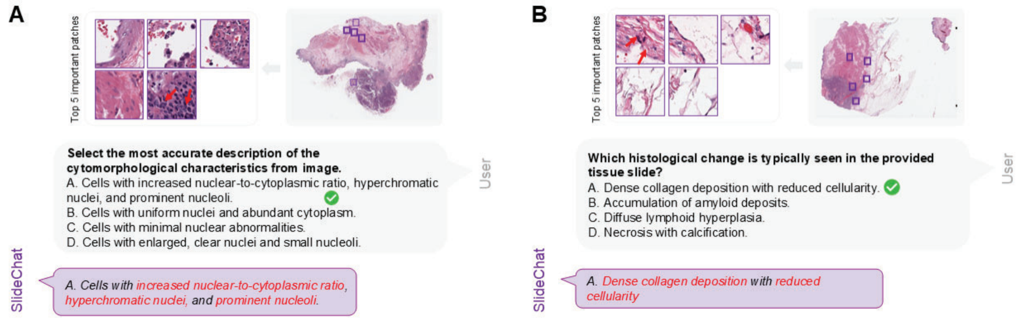

Figure 5. Interpretability and visualization. We identify the top five patch tokens with the highest attention scores associated with the output text responses.

likely due to the difficulty of learning the complex visual features of WSIs without the slide-level encoder. When using a frozen LLM for complex text generation, a simple projection proved inadequate for effectively integrating and aligning visual and textual features. Subsequently, we consider combining data from both stages and training SlideChat without the slide-level encoder by simultaneously updating both the projection layers and the LLM. Under this paradigm, performance on SlideBench-VQA (TCGA) and WSI-VQA*, which share the same distribution as the training set, was comparable to the two-stage training configuration with the slide-level encoder. However, a significant decline is observed when evaluating SlideBench-VQA (BCNB), which originates from a different domain; overall performance dropped by over 10% (Tab. 3), indicating a substantial reduction in the model's generalization ability. Therefore, we recommend incorporating a slide-level encoder to capture the complex visual features of whole slide, as it is particularly effective for cross-domain alignment and enhances the model's generalization performance.

Interpretability Despite SlideChat demonstrating promising results, concerns remained regarding the model's perception of large pathological slides. To further assess the model's interpretability, we calculated the correlation between the text output and specific image patches, thereby obtaining patch-level attention scores. By identifying the most significant patches, we gained insights into the precise areas the model focused on during response generation. Highlighting the most relevant regions of the tissue slides not only enhances transparency and bolsters the reliability of the model's outputs but also assists pathologists by directing attention to critical areas requiring closer scrutiny. Ultimately, such interpretability is essential for fostering trust in AI-assisted diagnostics and enhancing the precision and efficiency of clinical evaluations. As shown in Fig. 5, we are pleased to observe that the top five important patches identified by the model closely corresponded with the features described in SlideChat's output. Our extraction method retrieves attention weights for patch tokens from each generated token, averaging them across layers and heads. We then identify the top five patch tokens with the highest attention weights for further analysis. For example, in Fig. 5 (A), the highlighted patches clearly emphasized regions exhibiting an increased nuclear-to-cytoplasmic ratio, hyperchromatic nuclei, and prominent nucleoli. Similarly, in Fig. 5 (B), the selected patches demonstrated areas with dense collagen deposition and reduced cellularity, as detailed in the model's response. This alignment between the highlighted image regions and the textual outputs significantly enhances the model's interpretability, providing increased confidence that it accurately captures and assesses relevant histopathological features. Such consistency deepens our understanding of the model's reasoning processes regarding pathological slides and underscores the potential for integrating these AI systems into clinical workflows with greater assurance.

5. Conclusion

In this work, we present SlideChat, the first vision-language assistant capable of understanding gigapixel whole-slide images. Furthermore, we creat SlideInstruction, a largest comprehensive WSI instruction-following dataset to develop SlideChat, as well as SlideBench, a multi-modal benchmark designed to evaluate SlideChat across diverse scenarios. SlideChat demonstrates excellent chat abilities and achieves state-of-the-art performance on 18 tasks. We bridge the gap between MLLMs and pathology images at the whole-slide level with SlideChat, and we believe this represents an advancement in the field of general medical artificial intelligence (GMAI).

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